



YEAR 12 Mathematics

HSC Course

Assessment Task 2

March 2010

1. There are 3 questions.
2. Marks allocated to each question are indicated in brackets
3. Answer each question on your own paper showing all necessary working
4. Start each question on a new page
5. Calculators may be used
6. Time allowed - **90 minutes**

Topic	Mark
1. Question 1 (Geometrical Applications of Calculus)	/25
2. Question 2 (Integration)	/35
3. Question 3 (Trigonometric Functions)	/34

TOTAL

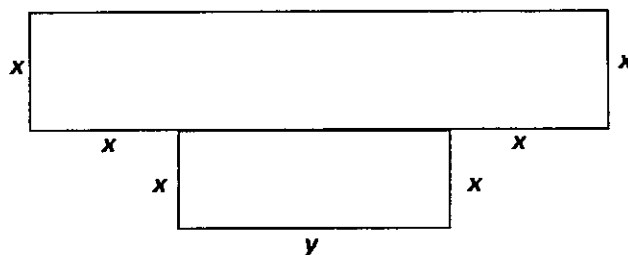
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Question 1 (25 marks)

a) Consider the curve $y = \frac{4x^3 - x^4}{9}$

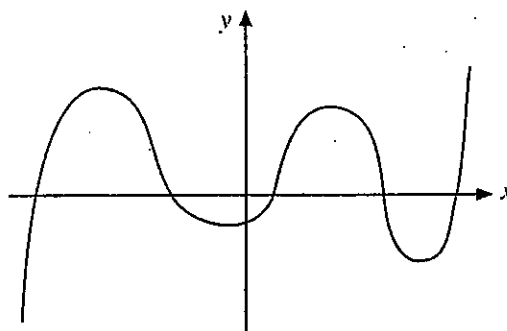
- I. Find any stationary points, and determine their nature (4)
- II. Find any points of inflexion (2)
- III. For what values of x is the curve concave up? (2)
- IV. Neatly sketch the curve in the domain $-3 \leq x \leq 4$ showing all important features (3)

- b) The enclosure below. In which all angles are right angles, is to be fenced with 120m of fencing. (all drawn lines are fences and measurements are all in metres)



- I. If the total area enclosed is $A \text{ m}^2$, show that $A = 2x^2 + 2xy$, and express this as a function of x alone. (3)
- II. Find for what value of x the enclosure has a maximum area (Justify your answer using calculus) (3)

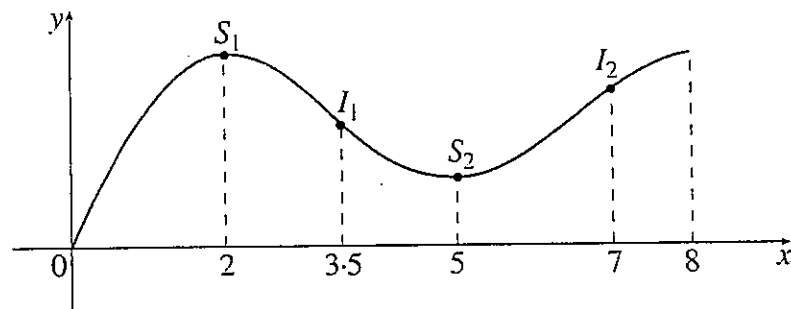
d)



The diagram above shows the graph of a function, $f(x)$.

- I. Copy the diagram and on the same diagram sketch and clearly label $y = f'(x)$ (3)

e)



The diagram above shows the graph of function $g(x)$ over the domain $0 \leq x \leq 8$. S_1 and S_2 are its two stationary points. I_1 and I_2 are its two points of inflection.

State all the intervals of x for which

I. $g'(x) > 0$ (3)

II. $g''(x) > 0$ (2)

Question 2 (35 marks)

a. Find

i. $\int (1 - 4x) dx$ (1)

ii. $\int (1 - 4x)^5 dx$ (2)

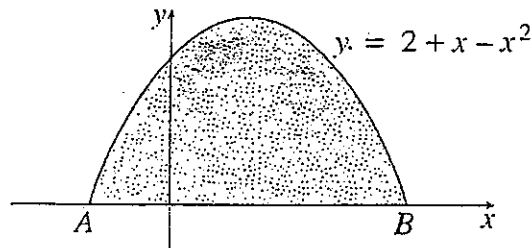
iii. $\int \frac{x^5 + x^2 + 1}{x^2} dx$ (2)

iv. $\int (1 - x^2)^2 dx$ (2)

b. Use the table of standard integrals to evaluate

$$\int_0^{\frac{\pi}{8}} \sec^2 2x dx$$
 (2)

c.



i. Find the coordinates of A and B (2)

ii. Find the area bounded by the curve and the x axis (3)

d. i. Copy this table and complete the missing values in decimal form.

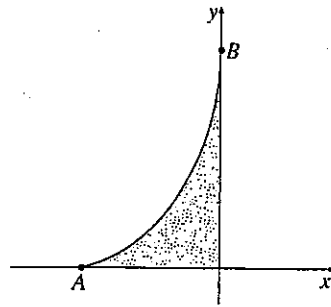
x	1	2	3	4	5
$\frac{3}{x(x+1)}$					

(2)

ii. Using all the above information, and Simpson's Rule, find an approximation to 2 decimal places for

$$\int_1^5 \frac{3}{x(x+1)} dx$$
 (2)

- e. The shaded region in the diagram below is bounded by the coordinate axes and part of the curve $y = (x+2)^3$.



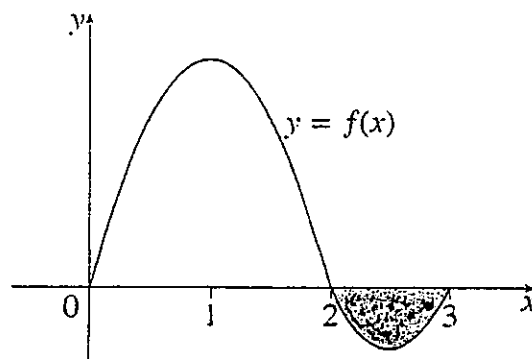
- What are the coordinates of A and B? (2)
- Find the volume of the solid formed when the region makes a revolution about the x axis. (3)
- If the region is revolved about the y axis, show that the volume, V , of the resulting solid is given by (2)

$$V = \pi \int_0^8 (y^{\frac{1}{3}} - 2)^2 dy$$

- f. Consider the curves $y = (x-1)^2$ and $y = 10x - x^2 - 9$

- Sketch the curves carefully on the same diagram, showing their point(s) of intersection (4)
- Find the area bounded by these two curves (4)

g.



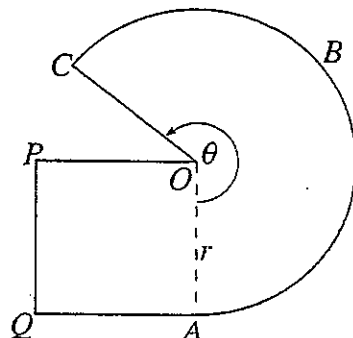
Given that $\int_0^2 f(x) dx = 11$ and that $\int_0^3 f(x) dx = 7$

- What is the area of the shaded region? (1)
- What is the value of $\int_2^3 f(x) dx$ (1)

Question 3 (34 marks)

- a. Convert 18° to radians, giving your answer in terms of π . (2)

b.



The diagram above shows a sector OABC of a circle, radius r , together with a square OPQA of side r .

Reflex $\angle AOC = \theta$ radians

- i. If the perimeter of the sector is equal to the circumference of the circle radius r , prove that $\theta = 2\pi - 2$ (3)
 - ii. Show that the total area of the figure is equal to that of a circle radius r . (3)
- c. Solve the following trigonometric equations for $0 \leq \theta \leq 2\pi$.
- i. $\cos 2\theta = -\frac{\sqrt{3}}{2}$ (3)
 - ii. $\sec^2 \theta + \tan \theta = 1$ (4)
- d. i. Sketch a neat graph of $y = 2\cos x$ for $-\pi \leq x \leq \pi$ (3)
- ii. The line $y = 1$ meets the curve in part i. at P and Q. Find the exact length of PQ. (3)
- e. Differentiate with respect to x
- i. $2\sin 3x$ (1)
 - ii. $\sin x \tan x$ (2)
 - iii. $\sqrt{\frac{x^2}{1 + \tan^2 x}}$ (3)
- f. For the function $y = 2\cos(2x - \frac{\pi}{4})$
- i. State the amplitude (1)
 - ii. State the period (2)
 - iii. Solve $2\cos(2x - \frac{\pi}{4}) = \sqrt{3}$, for $0 \leq x \leq \pi$ (4)

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE:

$$\ln x = \log_e x, x > 0$$

Q1/ (a) $y = \frac{4x^3 - x^4}{9}$
 $y' = \frac{12x^2 - 4x^3}{9}$

St. pts occur when $y' = 0$
 $12x^2 - 4x^3 = 0$
 $4x^2(3 - x) = 0$
 $x = 0$ and $x = 3$

$y'' = \frac{24x - 12x^2}{9}$

When $x = 0$, $y'' = 0$
 Test concavity either side

x	$0 - \epsilon$	0	$0 + \epsilon$
y''	$-$	0	$+$

 ∴ change in concavity

Horizontal POI at $(0, 0)$

When $x = 3$, $y'' < 0$ ∴ Max st pt at $(3, 3)$

(ii) Possible POI when $y'' = 0$

$24x - 12x^2 = 0$
 $12x(2 - x) = 0$
 $x = 0$ and $x = 2$

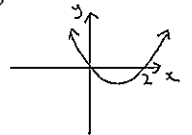
x	$2 - \epsilon$	2	$2 + \epsilon$
y''	$+$	0	$-$

 ∴ change in concavity

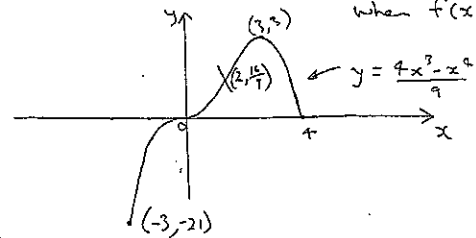
POI at $(2, \frac{16}{9})$ and $(0, 0)$ proven (ii)

(iii) Concave up when $y'' > 0$

$\frac{24x - 12x^2}{9} > 0$
 $0 > 12x^2 - 24x$
 $0 > 12x(x - 2)$



∴ $0 < x < 2$
 when $f(x)$ is concave up



(b) Fencing length = 120

Fencing = $4x + 2y + 2(2x + y)$
 $= 4x + 2y + 4x + 2y$
 $= 8x + 4y$ ∴ $8x + 4y = 120$

Area = $x(2x + y) + xy$
 $= 2x^2 + xy + xy$
 $= 2x^2 + 2xy$ but $8x + 4y = 120$
 $y = \frac{120 - 8x}{4}$

∴ $A = 2x^2 + 2x(\frac{120 - 8x}{4})$
 $= 2x^2 + 80x - \frac{16x^2}{4}$

Stat. values occur when $\frac{dA}{dx} = 0$
 (iii) $A = 2x^2 + 80x - \frac{16x^2}{4}$

$A = -\frac{10}{4}x^2 + 80x$

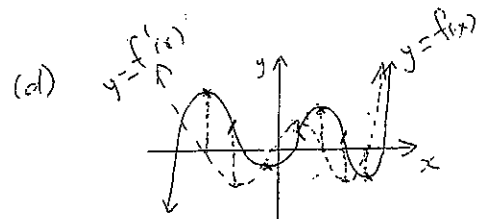
$\frac{dA}{dx} = -\frac{20}{4}x + 80$

$\frac{20x}{4} = 80$

$20x = 240$
 $x = 12$

$\frac{d^2A}{dx^2} = -\frac{20}{4}$ ∴ Concave up for all x

∴ Max area when $x = 12$



(e)(i) (Curve increasing) $0 \leq x < 2$, $5 \leq x \leq 8$

(ii) $3.5 < x < 7$
 (Curve concave up)

Q2/(a) (i) $x - 2x^2 + c$

(ii) $\frac{(1-4x)^6}{-24} + c$

(iii) $\int x^3 + 1 + x^{-2} dx = \frac{x^4}{4} + x - x^{-1} + c$
 $= \frac{x^4}{4} + x - \frac{1}{x} + c$

(iv) $\int 1 - 2x^2 + x^4 dx$
 $= x - \frac{2x^3}{3} + \frac{x^5}{5} + c$

(b) $\left[\frac{1}{2} \tan 2x \right]_0^{\frac{\pi}{8}}$
 $= \frac{1}{2} \tan \frac{\pi}{4} - \frac{1}{2} \tan 0$
 $= \frac{1}{2} (1) - \frac{1}{2} (0)$
 $= \frac{1}{2}$

(c) $y = 2 + x - x^2$
 (i) cuts x axis when $y = 0$

$x^2 - x - 2 = 0$
 $(x-2)(x+1) = 0 \quad \therefore A(-1, 0)$
 $x = 2 \text{ and } x = -1 \quad B(2, 0)$

(ii) $A = \int_{-1}^2 2 + x - x^2 dx$

$= \left[2x + \frac{x^2}{2} - \frac{x^3}{3} \right]_{-1}^2$
 $= 2(2) + \frac{(2)^2}{2} - \frac{(2)^3}{3} - \left[2(-1) + \frac{(-1)^2}{2} - \frac{(-1)^3}{3} \right]$
 $= 4 + 2 - \frac{8}{3} - \left[-2 + \frac{1}{2} + \frac{1}{3} \right]$
 $= 4\frac{1}{2} \text{ u}^2$

(d) (i)

x	1	2	3	4	5
$\frac{3}{x(x+1)}$	$\frac{3}{2}$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{3}{20}$	$\frac{1}{10}$
	1.5	0.5	0.25	0.15	0.1

(ii) $\frac{h}{3} (f_{\text{ist}} + f_{\text{ist}} + 2f_{\text{mid}} + 4f_{\text{em}})$
 $= \frac{1}{3} (1.5 + 0.1 + 2(0.25) + 4(0.5 + 0.15))$
 $= \cancel{2.5} \cdot 1.57 \text{ (to 2 dec places)}$

(e) $y = (x+2)^3$
 cuts y axis when $x = 0$
 $y = 8$

cuts x axis when $y = 0$
 $0 = (x+2)^3$
 $x = -2$

(i) $A(-2, 0)$ and $B(0, 8)$

(ii) $\pi \int_{-2}^0 y^2 dx = \pi \int_{-2}^0 (x+2)^6 dx$

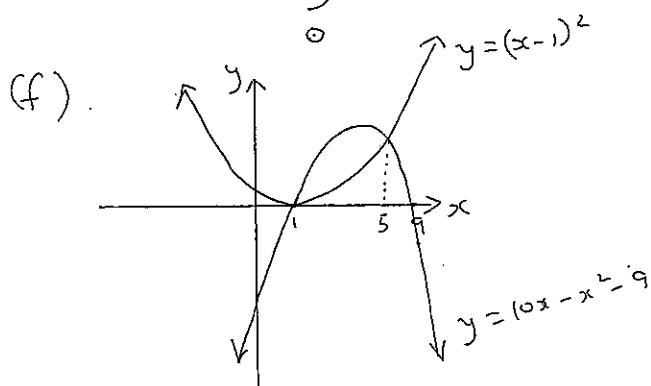
$= \frac{\pi}{7} \left[(x+2)^7 \right]_{-2}^0$
 $= \frac{\pi}{7} [2^7 - 0]$

(iii) $V = \pi \int_0^8 x^2 dy$

$y^{\frac{1}{3}} = x + 2$

$(y^{\frac{1}{3}} - 2) = x \quad (y^{\frac{1}{3}} - 2)^2 = x^2$

$V = \pi \int_0^8 (y^{\frac{1}{3}} - 2)^2 dy$



$y = -x^2 + 10x - 9$
 $y = -(x^2 - 10x + 9)$
 $= -(x-9)(x-1)$

$(x-1)^2 = 10x - x^2 - 9$
 $x^2 - 2x + 1 = 10x - x^2 - 9$

$2x^2 - 12x + 10 = 0$
 $x^2 - 6x + 5 = 0$
 $(x-5)(x-1) = 0$
 $x = 1 \text{ and } x = 5$

$A = \int_1^5 10x - x^2 - 9 - (x^2 - 2x + 1) dx$

$A = \int_1^5 10x - x^2 - 9 - x^2 + 2x - 1 dx$

$$= \int_1^5 -2x^2 + 12x - 10 \, dx$$

$$= \left[-\frac{2x^3}{3} + 6x^2 - 10x \right]_1^5$$

$$= \left[-\frac{2(5)^3}{3} + 6(5)^2 - 10(5) - \left(-\frac{2}{3} + 6 - 10 \right) \right]$$

$$= 21\frac{1}{3} \text{ u}^2$$

(g) (i) 4u^2
(ii) -4

Question - 3

(a) $180^\circ = \pi^c$

$$1^\circ = \frac{\pi}{180}$$

$$18 = \frac{18\pi}{180}$$

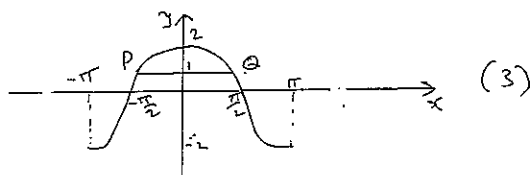
$$= \frac{\pi}{10}^c$$

(b) $C = 2\pi r$ Sector OABC = $r\theta + 2r$

$$\therefore 2\pi r = r\theta + 2r$$

$$2\pi = \theta + 2$$

(d)



(i) $2\cos x = 1$
 $\cos x = \frac{1}{2}$ $-\pi \leq x \leq \pi$
 $x = \pm \frac{\pi}{3}$

$$\therefore \text{distance PQ} = \frac{2\pi}{3}$$

(3)

(e) (i) $\frac{d}{dx} = 6 \cos 3x$ (1)

(ii) $\frac{d}{dx} = \sin x \cdot \sec^2 x + \tan x \cdot \cos x$ (2)
 $\frac{d}{dx} = \sin x \cdot \frac{1}{\cos^2 x} + \tan x \cdot \cos x$

$$\tan x \cdot \sec x + \tan x \cdot \cos x$$

$$\tan x (\sec x + \cos x)$$

(iii) $\frac{x}{\sec x} = x \cos x$

$$\frac{d}{dx} x \cos x = x \cdot -\sin x + \cos x \cdot 1$$

$$= -x \sin x + \cos x$$

$$= \cos x - x \sin x \quad (3)$$

$$2\pi r = r\theta + 2r$$

$$2\pi r - 2r = r\theta$$

$$r(2\pi - 2) = r\theta$$

$$\therefore \theta = 2\pi - 2$$

(ii) Area figure = $r^2 + \frac{1}{2}r^2\theta$

$$\text{Area circle} = \pi r^2$$

But Area figure = $r^2 + \frac{1}{2}r^2(2\pi - 2)$ from (i)

$$= r^2 + \pi r^2 - r^2$$

$$= \pi r^2$$

C. (i) $\cos 2\theta = -\frac{\sqrt{3}}{2}$ $0 \leq 2\theta \leq 2\pi$

$$\text{ref } \angle = \frac{\pi}{6}$$

$$2\theta = \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{17\pi}{6}, \frac{19\pi}{6}$$

$$\therefore \theta = \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{17\pi}{12}, \frac{19\pi}{12}$$

(ii) $1 + \tan^2 \theta + \tan \theta = 1$

$$\tan^2 \theta + \tan \theta = 0$$

$$\tan \theta (1 + \tan \theta) = 0$$

$$\tan \theta = 0 \quad \text{OR} \quad \tan \theta = -1$$

$$\theta = 0, \pi, 2\pi$$

$$\theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

(f) (i) $a = 2$

(ii) Period = $\frac{2\pi}{2}$

$$= \pi$$

(iii) $\cos(2x - \frac{\pi}{4}) = \frac{\sqrt{2}}{2}$ $0 \leq x \leq \pi$

$$0 \leq 2x \leq 2\pi$$

$$-\frac{\pi}{4} \leq 2x - \frac{\pi}{4} \leq 2\pi - \frac{\pi}{4}$$

$$\text{ref } \angle = \frac{\pi}{6}$$



$$2x - \frac{\pi}{4} = -\frac{\pi}{6}, \frac{\pi}{6}$$

$$2x = -\frac{\pi}{6} + \frac{\pi}{4}, \frac{\pi}{6} + \frac{\pi}{4}$$

$$2x = \frac{\pi}{12}, \frac{5\pi}{12}$$

$$x = \frac{\pi}{24}, \frac{5\pi}{24}$$